Extraction of E_{on}, E_{off} and E_{rr} from Double Pulse Test Waveforms

Application to a 1700 V/1200 A IGBT module (FF1200R17IP5)

Abstract

This note describes how to compute the switching energies E_{on}, E_{off} and E_{rr} from double pulse test (DPT) records consisting of the collector–emitter voltage $v_{\mathrm{CE}}(t)$ and the emitter current $i_{\mathrm{E}}(t)$ sampled at $T_s = 2\,\mathrm{ns}$. It covers the signal conditioning steps that dominate the final uncertainty (offset removal, probe deskew, stray inductance), the integration limits used by device manufacturers, the indirect determination of the diode reverse-recovery energy when only the DUT is instrumented, and the numerical implementation in MATLAB. The output is a set of energy values in the form required by a lookup-table loss model.

1 Test circuit and measured quantities

The standard DPT circuit is shown in Fig. 1. The device under test (DUT) is the low-side IGBT T_2; the load inductor L_{load} is connected between the phase output and the positive rail, so that the load current freewheels through the high-side diode D_1 when T_2 is off. The high-side IGBT T_1 is permanently off.

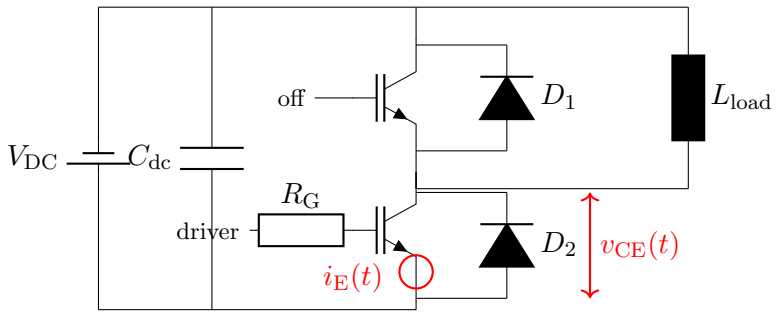


Figure 1: Double pulse test circuit. DUT: low-side IGBT T_2; commutation partner: high-side diode D_1.

The gate command consists of two pulses. The first, long pulse magnetises L_{load} up to the target current; the short off-interval between the two pulses produces the turn-off event; the second pulse produces the turn-on event. Only the *first turn-off* and the *second turn-on* are evaluated, because only for these two transitions is the current well defined and the device in a thermally and electrically settled state.

The two recorded channels are

$$v_{\mathrm{CE}}(t) \quad \text{measured at the module power terminals (C and E),} \tag{1}$$

$$i_{\mathrm{E}}(t) \quad \text{measured in the emitter lead.} \tag{2}$$

Since the gate current is at most a few amperes and flows only for tens of nanoseconds, $i_{\mathrm{E}}(t) \simeq i_{\mathrm{C}}(t)$ is assumed throughout; if a coaxial shunt in the emitter lead is used, the gate-current contribution can be subtracted if i_{G} is recorded.

2 Signal conditioning

The numerical integration itself is trivial. Essentially all the achievable accuracy is decided in the pre-processing.

2.1 Offset and baseline removal

Both channels must have zero mean over an interval in which the physical quantity is known to be zero:

$$\hat{\imath}(t) = i(t) - \frac{1}{N_0} \sum_{k \in \mathcal{W}_0} i(t_k), \tag{3}$$

where $\mathcal{W}_0$ is a pre-trigger window (typically 1 to 5 µs) taken before the first gate pulse, where the current is exactly zero. The same applies to v_{CE} only if a window exists where the device is fully on; otherwise the voltage offset is calibrated separately.

With $I_{\mathrm{L}} = 1200\,\mathrm{A}$ and a typical overlap duration of 300 ns, a current offset of 5 A already produces an error of the order of 1.4 mJ on E_{on}. Rogowski coils additionally suffer from integrator droop: the baseline must be estimated *immediately* before the transient, not at the beginning of the record.

2.2 Probe deskew

This is the dominant error source. Let δ be the residual time misalignment between the current and the voltage channel. The measured energy is

$$E(\delta) = \int v(t)\, i(t+\delta)\, \mathrm{d}t \simeq E(0) + \delta \int v(t)\, \frac{\mathrm{d}i}{\mathrm{d}t}\, \mathrm{d}t. \tag{4}$$

During turn-off the voltage is close to V_{DC} while the current falls from I_{L} to zero, hence $\int v\,\mathrm{d}i \simeq -V_{\mathrm{DC}} I_{\mathrm{L}}$ and

$$\boxed{\left|\frac{\Delta E}{E}\right| \simeq \frac{V_{\mathrm{DC}}\, I_{\mathrm{L}}}{E}\, |\delta| = \frac{|\delta|}{\tau_{\mathrm{eq}}}, \qquad \tau_{\mathrm{eq}} \triangleq \frac{E}{V_{\mathrm{DC}} I_{\mathrm{L}}}} \tag{5}$$

where τ_{eq} is the equivalent rectangular overlap time. For $V_{\mathrm{DC}} = 900\,\mathrm{V}$, $I_{\mathrm{L}} = 1200\,\mathrm{A}$ and $E \approx 300\,\mathrm{mJ}$ one gets $\tau_{\mathrm{eq}} \approx 278\,\mathrm{ns}$: a deskew error of 10 ns (five samples at $T_s = 2\,\mathrm{ns}$) causes a 3.6 % energy error, and the errors on E_{on} and E_{off} have *opposite* sign, so the sum is much less sensitive than the individual terms.

Two practical procedures:

1. **Hardware**: use a deskew fixture before the measurement campaign. This is the only reliable method and should always be preferred.

2. **Software**: exploit the fact that the instantaneous power dissipated in a passive device cannot be significantly negative. Define

$$J(\delta) = \int_{t_1}^{t_2} \big[\max\big(0,\, -v(t)\, i(t+\delta)\big)\big]^2 \mathrm{d}t, \qquad \delta^\star = \arg\min_{\delta} J(\delta), \tag{6}$$

   evaluated on the turn-on transient, where a skew produces a large spurious negative power lobe. Sub-sample resolution is obtained by interpolating i on a shifted grid. This is a correction, not a substitute for hardware deskew.

2.3 Bandwidth and sampling

At $T_s = 2\,\mathrm{ns}$ the Nyquist frequency is 250 MHz, adequate for a 1700 V Si IGBT whose transients contain little energy above 100 MHz. With $\mathrm{d}v/\mathrm{d}t = 5\,\mathrm{kV/\mu s}$ the voltage moves 10 V per sample and with $\mathrm{d}i/\mathrm{d}t = 5\,\mathrm{kA/\mu s}$ the current moves 10 A per sample: the trapezoidal rule then has a truncation error well below 0.1 %. A convergence check is obtained by decimating the record by factors 2, 4, 8 and verifying that the energy changes by less than the target tolerance.

2.4 Stray inductance

The voltage measured at the power terminals contains the drop across the internal inductance L_σ between the sensing points and the chip:

$$v_{\mathrm{CE}}^{\mathrm{term}}(t) = v_{\mathrm{chip}}(t) + L_\sigma \frac{\mathrm{d}i_{\mathrm{C}}}{\mathrm{d}t}. \tag{7}$$

Integrating the difference gives a closed-form bias:

$$E^{\mathrm{term}} - E^{\mathrm{chip}} = \int L_\sigma \frac{\mathrm{d}i_{\mathrm{C}}}{\mathrm{d}t}\, i_{\mathrm{C}}\, \mathrm{d}t = \tfrac{1}{2} L_\sigma \left(I_{\mathrm{end}}^2 - I_{\mathrm{start}}^2 \right), \tag{8}$$

i.e. $+\tfrac{1}{2} L_\sigma I_{\mathrm{L}}^2$ at turn-on and $-\tfrac{1}{2} L_\sigma I_{\mathrm{L}}^2$ at turn-off. With $L_\sigma = 15\,\mathrm{nH}$ and $I_{\mathrm{L}} = 1200\,\mathrm{A}$ this is 10.8 mJ per transition.

Two remarks:

- Manufacturers integrate the *terminal* quantities. If the objective is to reproduce or verify datasheet numbers, no de-embedding must be applied.
- The *external* commutation-loop inductance shifts energy between E_{on} and E_{off} (the rail collapse during turn-on lowers the measured E_{on}, the overvoltage at turn-off raises the measured E_{off}). The sum $E_{\mathrm{on}} + E_{\mathrm{off}} + E_{\mathrm{rr}}$ is therefore far more representative of the test bench than each individual term, and any comparison against datasheet values must be made at comparable loop inductance.

3 Definitions

The instantaneous dissipated power of the DUT is

$$p(t) = v_{\mathrm{CE}}(t)\, i_{\mathrm{C}}(t), \tag{9}$$

and the switching energy is its integral over an interval whose limits are conventional:

$$E = \int_{t_1}^{t_2} v_{\mathrm{CE}}(t)\, i_{\mathrm{C}}(t)\, \mathrm{d}t. \tag{10}$$

3.1 Operating point

Both integration limits and the reported operating point are referred to the two levels

$$V_{\mathrm{DC}} = \langle v_{\mathrm{CE}}(t) \rangle_{t \in \mathcal{W}_{\mathrm{off}}}, \qquad I_{\mathrm{L}} = \langle i_{\mathrm{C}}(t) \rangle_{t \in \mathcal{W}_{\mathrm{on}}}, \tag{11}$$

where $\mathcal{W}_{\mathrm{off}}$ is a window in the off-state well after the ringing has decayed (a few µs before the second turn-on) and $\mathcal{W}_{\mathrm{on}}$ is a window of 0.2 to 1 µs immediately before the turn-off command. Note that, because of the inductor magnetisation during the off-time and the DC-link droop, the current at turn-off and at turn-on differ slightly: each energy must be reported with *its own* current.

3.2 Integration limits

The conventional limits, consistent with IEC 60747-9 and with the definitions used in IGBT module datasheets, are summarised in Table 1 and illustrated in Fig. 2.

Table 1: Integration limits.

Energy	Start t_1	Stop t_2	Integrand
E_{on}	$i_{\mathrm{C}} \uparrow 0.10\, I_{\mathrm{L}}$	$v_{\mathrm{CE}} \downarrow 0.02\, V_{\mathrm{DC}}$	$v_{\mathrm{CE}} i_{\mathrm{C}}$
E_{off}	$v_{\mathrm{CE}} \uparrow 0.10\, V_{\mathrm{DC}}$	$i_{\mathrm{C}} \downarrow 0.02\, I_{\mathrm{L}}$	$v_{\mathrm{CE}} i_{\mathrm{C}}$
E_{rr}	$i_{\mathrm{D}} \downarrow 0$	$\lvert i_{\mathrm{D}} \rvert \downarrow 0.02\, I_{\mathrm{rrm}}$	$v_{\mathrm{D}} i_{\mathrm{D}}$

Formally, with $\alpha = 0.10$ and $\beta = 0.02$,

$$t_1^{\mathrm{on}} = \min\{t : i_{\mathrm{C}}(t) \geq \alpha I_{\mathrm{L}}\}, \qquad t_2^{\mathrm{on}} = \min\{t > t_1^{\mathrm{on}} : v_{\mathrm{CE}}(t) \leq \beta V_{\mathrm{DC}}\}, \tag{12}$$

$$t_1^{\mathrm{off}} = \min\{t : v_{\mathrm{CE}}(t) \geq \alpha V_{\mathrm{DC}}\}, \qquad t_2^{\mathrm{off}} = \min\{t > t_1^{\mathrm{off}} : i_{\mathrm{C}}(t) \leq \beta I_{\mathrm{L}}\}. \tag{13}$$

The crossing instants must be found by linear interpolation between samples, not by sample indexing, and the signals must be lightly filtered (or the crossing detected with hysteresis) so that ringing does not trigger a false crossing.

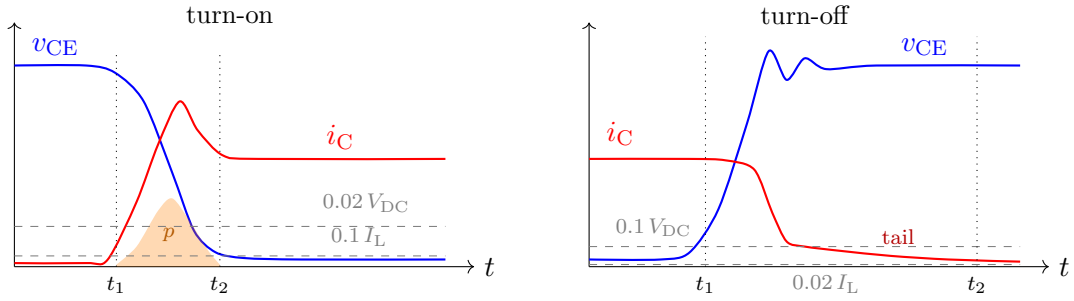


Figure 2: Integration limits for turn-on (left) and turn-off (right). The turn-on current peak is the reverse recovery of the opposite diode; the turn-off current tail is the recombination of the drift-region excess carriers.

3.3 The turn-off tail

For an IGBT the $\beta = 0.02$ criterion applied to the current is critical. At $I_{\mathrm{L}} = 1200\,\mathrm{A}$ the stop threshold is 24 A, and the tail current may still exceed this level for several microseconds while the full DC-link voltage is applied: a 20 A tail at 900 V for 2 µs carries 36 mJ, i.e. around 10 % of E_{off}. Two rules follow:

- the acquisition window after turn-off must be long enough (10 to 20 µs), otherwise the tail is truncated by the record, not by the criterion;
- the current channel must be accurate at the ampere level while ranged for 1200 A. This is the practical reason why the tail contribution is the least reproducible part of E_{off}.

It is good practice to report E_{off} together with its sensitivity, e.g. the value obtained with $\beta = 0.02$ and with a fixed 10 µs window.

4 Reverse recovery energy from the DUT waveforms

E_{rr} belongs to the diode D_1, which is normally not instrumented. It can be reconstructed from the DUT channels because the commutation loop imposes two algebraic relations. Taking the diode current i_{D} positive in the forward direction and the diode voltage v_{D} as anode-to-cathode:

$$i_{\mathrm{D}}(t) = I_{\mathrm{L}} - i_{\mathrm{C}}(t), \tag{14}$$

$$v_{\mathrm{D}}(t) = v_{\mathrm{CE}}(t) - V_{\mathrm{DC}} + L_{\sigma,\mathrm{dc}} \frac{\mathrm{d}i_{\mathrm{C}}}{\mathrm{d}t} \;\simeq\; v_{\mathrm{CE}}(t) - V_{\mathrm{DC}}. \tag{15}$$

Therefore

$$\boxed{E_{\mathrm{rr}} = \int_{t_1^{\mathrm{rr}}}^{t_2^{\mathrm{rr}}} \big(V_{\mathrm{DC}} - v_{\mathrm{CE}}(t)\big)\,\big(i_{\mathrm{C}}(t) - I_{\mathrm{L}}\big)\,\mathrm{d}t} \tag{16}$$

which is non-negative during recovery, since there $i_{\mathrm{C}} > I_{\mathrm{L}}$ and $v_{\mathrm{CE}} < V_{\mathrm{DC}}$. The limits are

$$t_1^{\mathrm{rr}} = \min\{t :\; i_{\mathrm{C}}(t) \geq I_{\mathrm{L}}\} \quad \text{(diode current zero crossing)}, \tag{17}$$

$$t_2^{\mathrm{rr}} = \min\{t > t_{\mathrm{peak}} :\; i_{\mathrm{C}}(t) - I_{\mathrm{L}} \leq 0.02\, I_{\mathrm{rrm}}\}, \qquad I_{\mathrm{rrm}} = \max_t \big(i_{\mathrm{C}}(t) - I_{\mathrm{L}}\big). \tag{18}$$

The recovery charge, useful as a cross-check and for parameterising a diode model, is

$$Q_{\mathrm{rr}} = \int_{t_1^{\mathrm{rr}}}^{t_2^{\mathrm{rr}}} \big(i_{\mathrm{C}}(t) - I_{\mathrm{L}}\big)\,\mathrm{d}t, \qquad \text{with the consistency check } E_{\mathrm{rr}} \approx k\, Q_{\mathrm{rr}} V_{\mathrm{DC}},\; k \in [0.4, 0.6]. \tag{19}$$

Limitations of the indirect method, to be stated explicitly whenever it is used:

- V_{DC} is assumed constant, whereas the rail collapses by $L_{\sigma,\mathrm{dc}}\,\mathrm{d}i/\mathrm{d}t$ during the current rise. If a probe is available on the DC-link close to the module, V_{DC} should be replaced by the measured instantaneous rail voltage, and (15) becomes exact up to the residual layout inductance.
- The inductance between the two chips is attributed entirely to one device, so the split between E_{on} and E_{rr} is affected while $E_{\mathrm{on}} + E_{\mathrm{rr}}$ is not.
- The diode forward conduction before the zero crossing is excluded by construction, which is consistent with the datasheet definition of E_{rr}.

Note also that E_{on} of the IGBT computed as in Table 1 *already includes* the effect of the recovery current, because the recovery peak flows through the IGBT: E_{on} and E_{rr} must not be corrected for double counting, they are two distinct energies dissipated in two distinct chips.

5 Numerical implementation

5.1 Crossing detection

Given a signal x sampled on a uniform grid and a threshold X, the crossing instant is obtained by linear interpolation between the two samples that bracket it:

$$t^\star = t_k + T_s\,\frac{X - x_k}{x_{k+1} - x_k}, \qquad k = \min\{j :\; (x_j - X)(x_{j+1} - X) \leq 0\}. \tag{20}$$

The integral over $[t_1, t_2]$ with non-grid endpoints is computed with the trapezoidal rule on the interior samples plus the two partial trapezoids at the ends; the values $p(t_1)$ and $p(t_2)$ are themselves obtained by linear interpolation.

5.2 MATLAB code

Listing 1: Utility functions: threshold crossing and interpolated integral.

```
function tc = tcross(t, x, level, dir, tstart)
% First instant after TSTART at which X crosses LEVEL in direction DIR (+1/-1).
    if nargin < 5, tstart = t(1); end
    i0 = find(t >= tstart, 1, 'first');
    s = dir*(x(i0:end) - level);
    k = find(s(1:end-1) < 0 & s(2:end) >= 0, 1, 'first');
    if isempty(k), tc = NaN; return; end
    k = k + i0 - 1;
    tc = t(k) + (t(k+1)-t(k))*(level-x(k))/(x(k+1)-x(k));
end

function E = trapz_interval(t, p, t1, t2)
% Trapezoidal integral of P(t) between arbitrary instants T1 and T2.
    m = t > t1 & t < t2;
    p1 = interp1(t, p, t1); p2 = interp1(t, p, t2);
    tt = [t1; t(m); t2];
    pp = [p1; p(m); p2];
    E = trapz(tt, pp);
end
```

Listing 2: Pre-processing: offset removal and deskew.

```
function [t, v, i] = dpt_prepare(t, v, i, w0, delta)
% W0 : [t_start t_stop] pre-trigger window with i == 0
% DELTA : current-channel delay to be compensated [s] (i is shifted by -DELTA)
    t = t(:); v = v(:); i = i(:);
    m = t >= w0(1) & t <= w0(2);
    i = i - mean(i(m)); % current offset
    if nargin >= 5 && delta ~= 0
        i = interp1(t, i, t + delta, 'pchip', 'extrap');
    end
end
```

Listing 3: Operating point and switching energies.

```
function S = dpt_energies(t, v, i, tOff, tOn, opt)
% TOFF, TON : approximate instants of the turn-off and turn-on commands [s]
% OPT.alpha (0.10), OPT.beta (0.02), OPT.wOn, OPT.wOff : averaging windows [s]

    if ~isfield(opt,'alpha'), opt.alpha = 0.10; end
    if ~isfield(opt,'beta'), opt.beta = 0.02; end
    if ~isfield(opt,'wOn'), opt.wOn = 0.5e-6; end % before turn-off
    if ~isfield(opt,'wOff'), opt.wOff = 2.0e-6; end % before turn-on

    % ---- operating point ------------------------------------------------
    mI = t > tOff - opt.wOn & t < tOff - 50e-9;
    mV = t > tOn - opt.wOff & t < tOn - 50e-9;
    IL = mean(i(mI)); % load current at turn-off
    Vdc = mean(v(mV)); % blocking voltage before turn-on

    p = v.*i; % instantaneous power [W]

    % ---- turn-off -------------------------------------------------------
    t1 = tcross(t, v, opt.alpha*Vdc, +1, tOff - 100e-9);
    t2 = tcross(t, i, opt.beta *IL, -1, t1);
    S.Eoff = trapz_interval(t, p, t1, t2);
    S.toff = [t1 t2];

    % ---- turn-on --------------------------------------------------------
```

```
    ILon = mean(i(t > tOn + 3e-6 & t < tOn + 4e-6)); % current after turn-on
    t3 = tcross(t, i, opt.alpha*ILon, +1, tOn - 100e-9);
    t4 = tcross(t, v, opt.beta *Vdc, -1, t3);
    S.Eon = trapz_interval(t, p, t3, t4);
    S.ton = [t3 t4];

    % ---- reverse recovery of the opposite diode --------------------------
    irr = i - ILon; % diode current = ILon - i
    vrr = Vdc - v; % diode reverse voltage
    t5 = tcross(t, i, ILon, +1, t3); % diode current zero crossing
    mpk = t > t5 & t < t5 + 2e-6;
    [Irrm, kpk] = max(irr(mpk));
    idx = find(mpk, 1, 'first') + kpk - 1;
    t6 = tcross(t, irr, opt.beta*Irrm, -1, t(idx));
    S.Err = trapz_interval(t, vrr.*irr, t5, t6);
    S.Qrr = trapz_interval(t, irr, t5, t6);
    S.Irrm = Irrm;
    S.trr = [t5 t6];

    % ---- operating point and checks --------------------------------------
    S.IL = IL; S.ILon = ILon; S.Vdc = Vdc;
    S.Etot = S.Eon + S.Eoff + S.Err;
    S.k = S.Err/(S.Qrr*Vdc); % expected 0.4 ... 0.6
end
```

Listing 4: Software deskew estimate based on the non-negative power criterion.

```
function d = dpt_deskew(t, v, i, tOn, span)
% Minimises the spurious negative-power lobe on the turn-on transient.
    if nargin < 5, span = 1e-6; end
    m = t > tOn - 0.2e-6 & t < tOn + span;
    dd = (-20:0.25:20)*1e-9; % +/- 20 ns, 0.25 ns resolution
    J = zeros(size(dd));
    for k = 1:numel(dd)
        is = interp1(t, i, t + dd(k), 'pchip', 'extrap');
        p = v(m).*is(m);
        J(k) = trapz(t(m), max(0,-p).^2);
    end
    [~,k0] = min(J); d = dd(k0);
end
```

5.3 Convergence and sensitivity checks

Every extracted value should be accompanied by the following checks:

1. **Decimation**: recompute with T_s, $2T_s$, $4T_s$; the variation must be below $0.5\,\%$.
2. **Threshold sensitivity**: recompute with $(\alpha, \beta) = (0.10, 0.02)$, $(0.05, 0.01)$ and $(0.10, 0.05)$; the spread is a realistic estimate of the definition uncertainty and is typically a few percent for E_{on} and up to $10\,\%$ for E_{off} because of the tail.
3. **Deskew sensitivity**: recompute with $\delta \pm 2\,\mathrm{ns}$ and apply (5).
4. **Charge balance**: Q_{rr} from the current integral must be consistent with the datasheet at the same $\mathrm{d}i/\mathrm{d}t$ and T_{vj}.
5. **Energy balance**: $E_{\mathrm{on}} + E_{\mathrm{off}} + E_{\mathrm{rr}}$ is the quantity that should be compared across benches; individual terms depend on the loop inductance.

6 From single events to a loss model

A DPT campaign sweeps the current and, if the bench allows it, the voltage, the gate resistance and the junction temperature. The resulting data set is condensed as follows.

6.1 Current dependence

For a given $(V_{\mathrm{DC}}, R_{\mathrm{G}}, T_{\mathrm{vj}})$ the energies are well represented by a second-order polynomial through the measured points:

$$E(I) = a_0 + a_1 I + a_2 I^2. \tag{21}$$

a_0 absorbs the voltage-dependent, current-independent part (displacement currents, recovery of the residual charge); imposing $a_0 = 0$ is not recommended because it biases the fit at high current, which is where the loss actually matters.

6.2 Voltage dependence

Over a limited range around the reference voltage a power law is adequate:

$$E(V_{\mathrm{DC}}) = E_{\mathrm{ref}} \left(\frac{V_{\mathrm{DC}}}{V_{\mathrm{ref}}} \right)^{K_v}, \tag{22}$$

with K_v identified from at least three voltage levels. Typical values for Si IGBT modules are $K_v \approx 1.2 \ldots 1.4$ for E_{on} and E_{rr} and $K_v \approx 1.0 \ldots 1.2$ for E_{off}; a linear scaling ($K_v = 1$) is the usual first approximation and underestimates E_{on} at high voltage.

6.3 Temperature dependence

Linear interpolation between two characterised temperatures is standard:

$$E(T_{\mathrm{vj}}) = E_{25} + \big(E_{150} - E_{25}\big) \frac{T_{\mathrm{vj}} - 25}{125}. \tag{23}$$

The temperature coefficient is small for E_{on}, moderate for E_{off} (longer tail) and large for E_{rr} (higher carrier lifetime and stored charge).

6.4 Table for a lookup-table loss model

The final deliverable is a table in the form of Table 2, one block per $(V_{\mathrm{DC}}, R_{\mathrm{G}}, T_{\mathrm{vj}})$ combination, directly usable as breakpoints in a Simscape or Simulink loss model. Reported currents must be the measured I_{L} values, not the nominal set points.

Table 2: Output format of the extraction (values to be filled with the measured data).

I_{L} (A)	V_{DC} (V)	T_{vj} (°C)	E_{on} (mJ)	E_{off} (mJ)	E_{rr} (mJ)	Q_{rr} (µC)
300	900	150				
600	900	150				
900	900	150				
1200	900	150				
1800	900	150				

7 Checklist

1. Verify the record contains a sufficient pre-trigger interval and at least 10 µs after turn-off.
2. Remove the current offset on the window immediately preceding the transient.
3. Apply the hardware deskew; verify with (6).
4. Extract V_{DC} and I_{L} (separately for the turn-off and turn-on events).
5. Locate t_1, t_2 by interpolated threshold crossing with $\alpha = 0.10$, $\beta = 0.02$.
6. Integrate $p = v_{\mathrm{CE}} i_{\mathrm{C}}$ for E_{on} and E_{off}; integrate $(V_{\mathrm{DC}} - v_{\mathrm{CE}})(i_{\mathrm{C}} - I_{\mathrm{L}})$ for E_{rr}.
7. Run the convergence and sensitivity checks; report the spread together with the value.
8. Compare $E_{\mathrm{on}} + E_{\mathrm{off}} + E_{\mathrm{rr}}$, not the individual terms, when validating against the datasheet, and only at comparable R_{G}, L_σ and T_{vj}.

For 1700 V modules the reference DC-link voltage used by manufacturers for the datasheet switching energies is 900 V, and E_{on} and E_{rr} refer to the commutation between the DUT and the diode of the opposite switch, so that the two energies share the same $\mathrm{d}i/\mathrm{d}t$ only if the gate resistances are the ones specified in the datasheet.

References

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[5] P. O. Lauritzen and C. L. Ma, "A simple diode model with reverse recovery," *IEEE Trans. Power Electron.*, vol. 6, no. 2, pp. 188–191, Apr. 1991.